FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects
   - If-Else statements

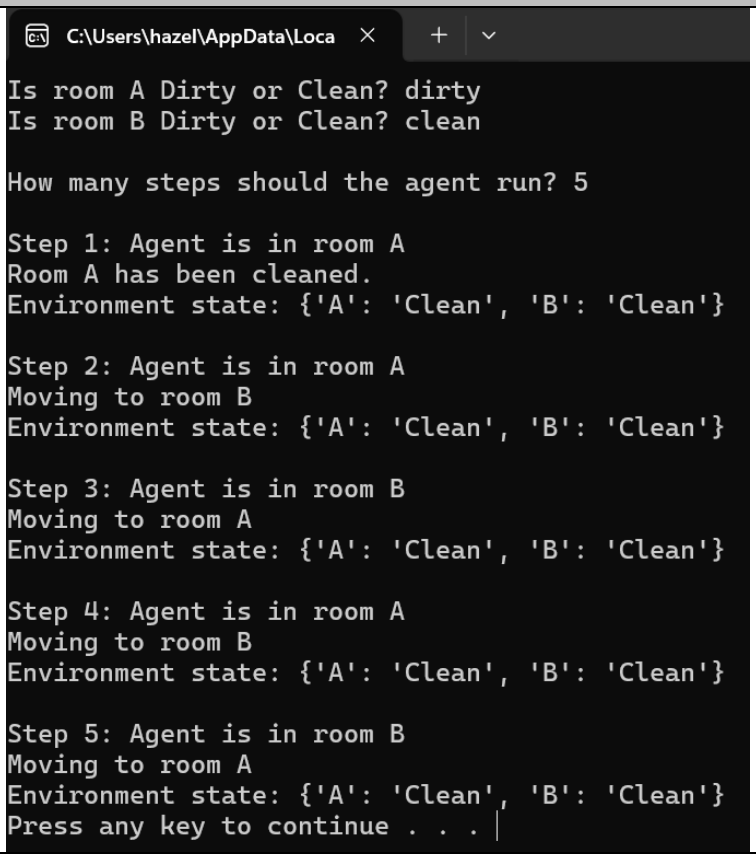
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

Task 3: Run the Simulation

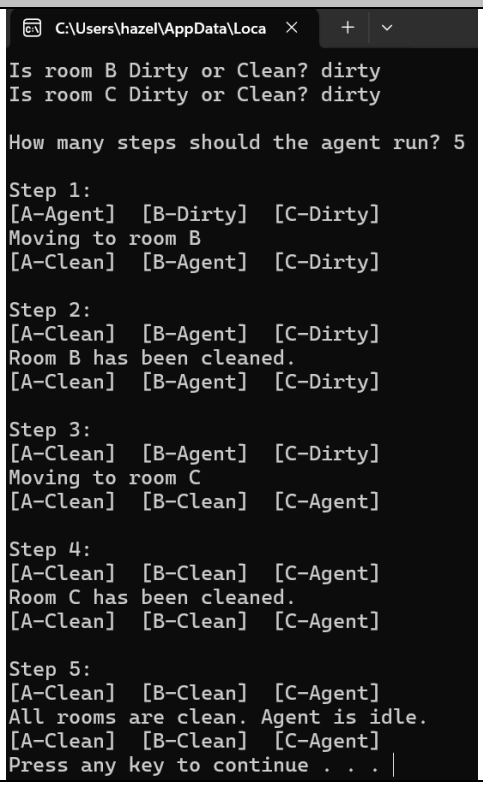
1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.

3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

CODE:

```
# dictionary
rooms = {
    "A": "clean",
    "B": "clean"
}

# check state
```

```
def is_dirty(room_state):
    if room_state == "dirty":
        return True
    else:
        return False

def clean_room(room_name):
    rooms[room_name] = "clean"

# rule-based agent class
class VacuumAgent:
    def __init__(self, start_location):
        self.location = start_location

    def move(self):

        if self.location == "A":
            self.location = "B"
        else:
            self.location = "A"

    def perceive_and_act(self):
        current_state = rooms[self.location]

        # Rule 1: if dirty, clean
        if is_dirty(current_state):
            clean_room(self.location)
            return "cleaned"

        # Rule 2: if clean, move
        else:
            self.move()
            return "moved"


print("Set the initial states of the rooms (type 'dirty' or 'clean').")
rooms["A"] = input("Room A state: ").lower()
rooms["B"] = input("Room B state: ").lower()

steps = int(input("How many steps should the simulation run? "))

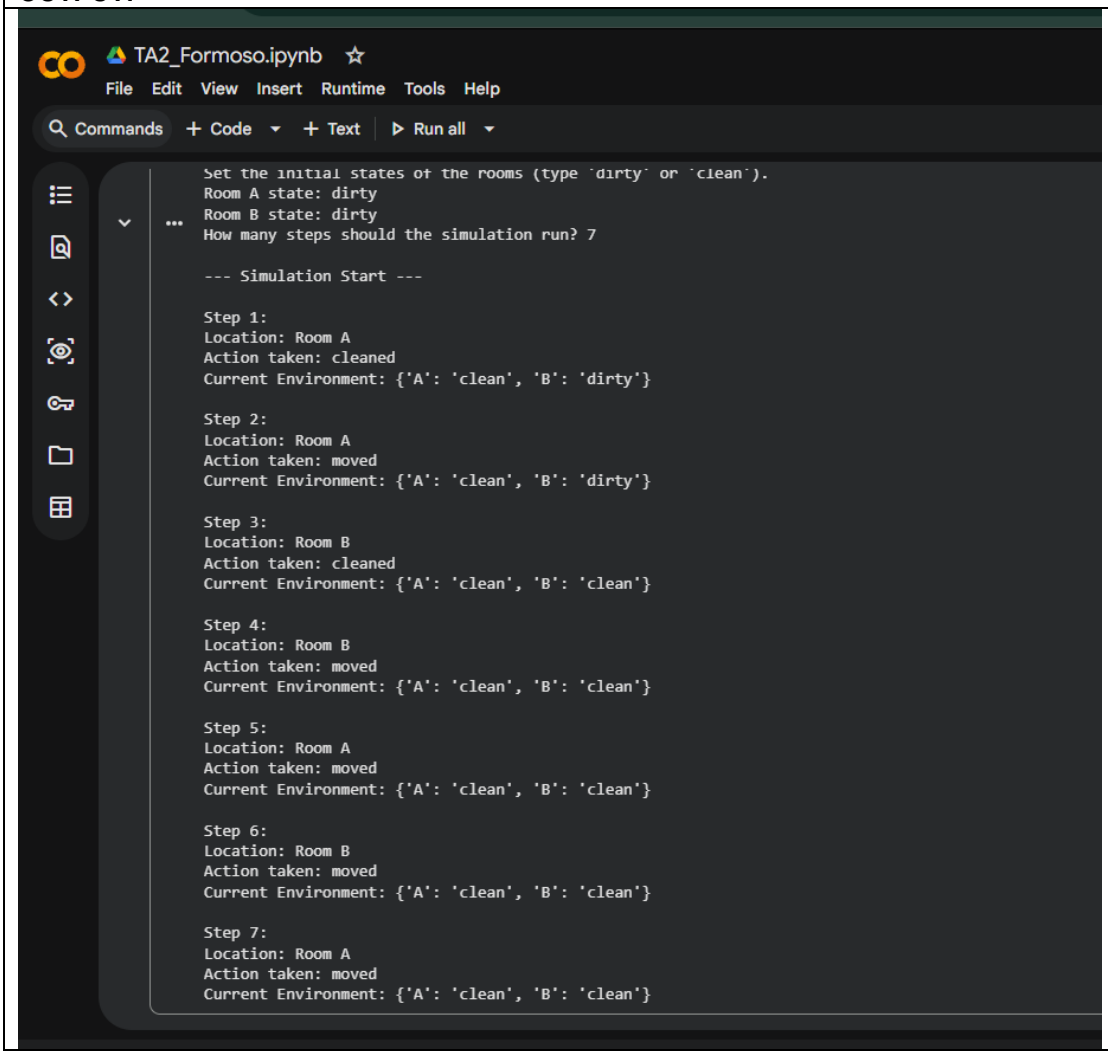
# start at A
agent = VacuumAgent("A")
```

```
print("\n--- Simulation Start ---")
for i in range(steps):
    print(f"\nStep {i + 1}:")
    print(f"Location: Room {agent.location}")

    action = agent.perceive_and_act()

    print(f"Action taken: {action}")
    print(f"Current Environment: {rooms}")
```

OUTPUT:



Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
   - It follows two basic instructions outlined in the lab: if the room it is currently in is dirty, it cleans it. If the current room is already clean, it moves over to the other room.
2. How does the agent decide when to clean and when to move?
   - The agent checks the dictionary value of its current location using the is_dirty() function. If it returns True, it runs the cleaning code. If it returns False, the script drops into the else block and calls the move() method to switch rooms instead.
3. What happens if both rooms are already clean at the start?
   - It will just keep bouncing back and forth indefinitely. Every step will just log a "moved" action since it never encounters a "dirty" state to trigger the cleaning rule.
4. How would you modify the agent to handle **more than two rooms**?
   - Add the new rooms (like room C) to the rooms dictionary. Then, change the move() method. Instead of just flipping between A and B, put the room keys into a list and make the agent cycle to the next index in the list, looping back to the first room when it hits the end.

Answer the Questions ⇩